$$= \left(\sqrt{\frac{537 \times 742}{442 \times 613}} \right) \times 100 = 121.2684$$

Time Reversal Test: $P_{01} \times P_{10} = 1$

$$P_{01} \times P_{10} = \sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum p_0q_1 \times \sum p_0q_0}{\sum p_0q_0 \times \sum p_0q_1 \times \sum p_1q_1 \times \sum p_1q_0} \right)}$$

$$P_{01} \times P_{10} = \sqrt{\left(\frac{537 \times 742 \times 613 \times 442}{442 \times 613 \times 742 \times 537} \right)}$$

$$P_{01} \times P_{10} = 1$$

Factor Reversal Test : $P_{01} \times Q_{01} = \dfrac{\sum p_1q_1}{\sum p_0q_0}$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum q_1p_0 \times \sum q_1p_1}{\sum p_0q_0 \times \sum p_0q_1 \times \sum q_0p_0 \times \sum q_0p_1} \right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{537 \times 742 \times 613 \times 742}{442 \times 613 \times 442 \times 537} \right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{742 \times 742}{442 \times 442} \right)} = \frac{742}{442}$$

$$\Rightarrow \quad P_{01} \times Q_{01} = \frac{\sum p_1q_1}{\sum p_0q_0}$$

Example 9.13

Calculate Fisher's price index number and show that it satisfies both Time Reversal Test and Factor Reversal Test for data given below.

Commodities	Base Year		Current Year	
	Price	Quantity	Price	Quantity
Rice	10	5	11	6
Wheat	12	6	13	4
Rent	14	8	15	7
Fuel	16	9	17	8
Transport	18	7	19	5
Miscellaneous	20	4	21	3

Solution:

Commodities	Base Year		Current Year		p_0q_0	p_0q_1	p_1q_0	p_1q_1
	Price (p_0)	Quantity (q_0)	Price (p_1)	Quantity (q_1)				
Rice	10	5	11	6	50	60	55	66
Wheat	12	6	13	4	72	48	78	52
Rent	14	8	15	7	112	98	120	105
Fuel	16	9	17	8	144	128	153	136
Transport	18	7	19	5	126	90	133	95
Miscellaneous	20	4	21	3	80	60	84	63
				Total	584	484	623	517

Table 9.12

Fisher's price index number

$$P_{01}^F = \left(\sqrt{\frac{\sum p_1q_0 \times \sum p_1q_1}{\sum p_0q_0 \times \sum p_0q_1}} \right) \times 100$$

$$= \left(\sqrt{\frac{623 \times 517}{584 \times 484}} \right) \times 100 = 106.74$$

Time Reversal Test: $P_{01} \times P_{10} = 1$

$$P_{01} \times P_{10} = \sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum p_0q_1 \times \sum p_0q_0}{\sum p_0q_0 \times \sum p_0q_1 \times \sum p_1q_1 \times \sum p_1q_0} \right)}$$

$$P_{01} \times P_{10} = \sqrt{\left(\frac{623 \times 517 \times 484 \times 584}{584 \times 484 \times 517 \times 623} \right)}$$

$$P_{01} \times P_{10} = 1$$

Factor Reversal Test : $P_{01} \times Q_{01} = \dfrac{\sum p_1q_1}{\sum p_0q_0}$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum q_1p_0 \times \sum q_1p_1}{\sum p_0q_0 \times \sum p_0q_1 \times \sum q_0p_0 \times \sum q_0p_1} \right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{623 \times 517 \times 484 \times 517}{584 \times 484 \times 584 \times 623} \right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{517 \times 517}{585 \times 584} \right)} = \frac{517}{584}$$

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